

Spike-timing-dependent plasticity

Spike-timing-dependent plasticity (STDP) is a biological process that adjusts the strength of synaptic connections between neurons based on the relative timing of their action potentials (or spikes). It is a temporally sensitive form of synaptic plasticity, meaning that the efficiency of synaptic transmission is modified by the timing of neural activity. When a presynaptic neuron consistently fires just before a postsynaptic neuron, the connection is typically strengthened—a process known as long-term potentiation (LTP). If the timing is reversed and the presynaptic neuron fires after the postsynaptic neuron, the connection is weakened through long-term depression (LTD).^{[1][2]}

STDP is considered a key mechanism in learning and memory formation and helps explain activity-dependent development of neural circuits. It has been observed in multiple brain regions, including the hippocampus, neocortex, and visual system, and has been widely implemented in computational models of biologically inspired learning algorithms and network dynamics.^{[3][4]}

STDP develops early in life, helping to refine sensory maps and establish functional connectivity during critical periods. The process depends on molecular mechanisms such as NMDA receptor-mediated calcium signaling^[5] and is influenced by synapse location and neuromodulators like dopamine and acetylcholine.^[6]

Variants of STDP have been found at inhibitory synapses and in response to complex spike patterns.^{[7][4]} The process also interacts with other forms of plasticity, including rate-based learning, homeostatic regulation, and structural remodeling.^{[8][9][10]} Disruptions in STDP have been linked to neurological and psychiatric conditions such as Alzheimer's disease, Fragile X syndrome, epilepsy, and Parkinson's disease.^{[11][12][13][14]}

Mechanism

Spike-timing-dependent plasticity (STDP) depends on the precise timing of action potentials (spikes) between presynaptic and postsynaptic neurons, hence the name: "spike-timing-dependent plasticity". If a presynaptic spike occurs shortly before a postsynaptic spike—typically within a window of 10 to 20 milliseconds—the synapse is strengthened, a process known as long-term potentiation (LTP). If the presynaptic spike follows the postsynaptic spike, the synapse is weakened, resulting in long-term depression (LTD).^{[1][2]} This timing-dependent adjustment of synaptic strength enables neurons to reinforce inputs that are likely to have contributed to their activation while weakening those that were not causally involved.^[3]

The effect of STDP is cumulative: repeated pairings of causally timed spikes strengthen the relevant synapses, while others weaken over time. Eventually, this leads to a subset of inputs being selectively retained, particularly those that tend to fire together within narrow temporal windows. As a result, the neuron becomes tuned to detect and respond preferentially to input patterns that consistently precede its own activation, which may reflect meaningful or predictive features of the environment.^[4]

At the molecular level, STDP is primarily mediated by N-methyl-D-aspartate receptors (NMDA receptors) located on the postsynaptic membrane. These receptors function as coincidence detectors: they require both the release of glutamate from the presynaptic terminal and sufficient depolarization of the postsynaptic membrane to become fully activated. When these conditions are met—such as when a back-propagating action potential follows synaptic input—the NMDA receptor channel opens, allowing calcium ions to enter the postsynaptic cell.^{[2][15]}

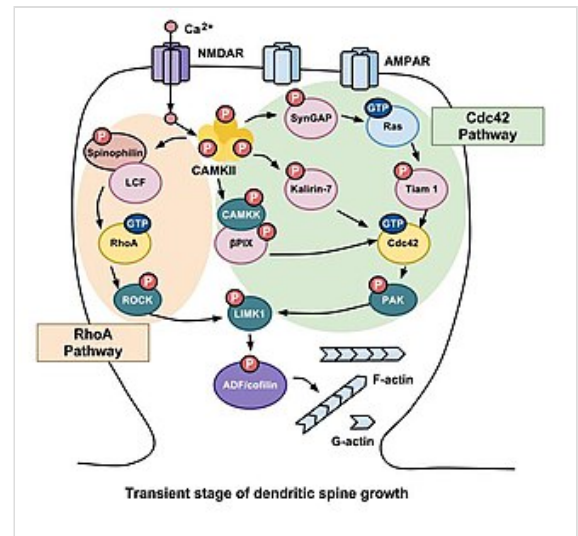
The amplitude and duration of calcium influx determine the direction of synaptic change. High-amplitude, rapid calcium transients typically trigger LTP via the activation of calcium-sensitive kinases, while lower, prolonged calcium levels are associated with LTD, in part due to the activation of phosphatases.^{[2][16]} The spike-timing rule is therefore shaped by intracellular signaling cascades that translate calcium signals into long-term structural or biochemical changes at the synapse.

Although the mechanisms of LTP are relatively well understood, the pathways underlying spike-timing-dependent LTD can vary. LTD may involve voltage-dependent calcium entry through other channels, activation of metabotropic glutamate receptors, or the release of retrograde messengers such as endocannabinoids. In some synapses, presynaptic NMDA receptors also contribute to LTD by modulating neurotransmitter release.^[16]

The specific shape of the STDP learning window—the curve that relates spike timing difference to synaptic change—differs across brain regions and cell types. Many synapses exhibit an asymmetric window favoring LTP for pre-before-post timing and LTD for post-before-pre. However, other synapses display symmetric, anti-Hebbian, or frequency-dependent patterns, particularly under different neuromodulatory conditions or in inhibitory circuits.^{[1][4][17]}

STDP in Inhibitory Synapses

Most early STDP research focused on excitatory (glutamatergic) synapses, but timing-dependent plasticity also occurs at inhibitory synapses. However, the rules of STDP at GABAergic synapses can differ significantly from their excitatory counterparts. In many cases, inhibitory STDP is anti-Hebbian or weight-dependent in the opposite manner. For example, in cortical circuits, fast-spiking parvalbumin-positive interneurons that synapse onto pyramidal cells exhibit timing-dependent changes in synaptic strength. When the interneuron fires slightly before the postsynaptic pyramidal neuron, providing feed-forward inhibition, the inhibitory synapse is weakened, a process known as **inhibitory long-term depression**. In contrast, if the interneuron fires after, or with a longer delay relative to the pyramidal neuron, the synapse is strengthened, resulting in inhibitory **long-term potentiation**.^{[7][18]}



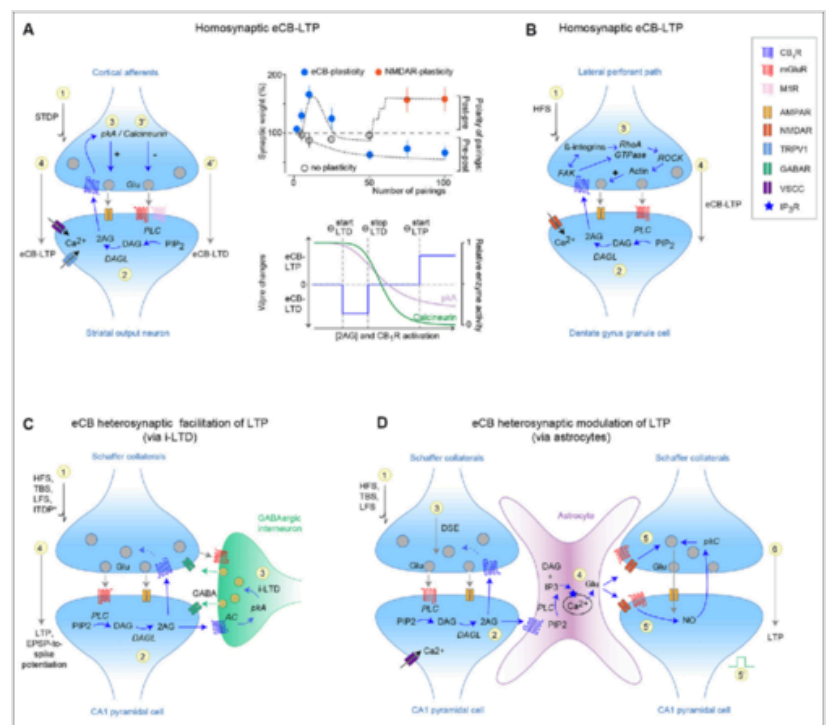
Calcium influx through NMDA receptors activates CaMKII, which initiates signaling cascades involving Rho family GTPases, primarily RhoA and Cdc42. These pathways regulate actin-binding proteins such as cofilin and profilin. During the transient phase, LIMK1 phosphorylates ADF/cofilin, inhibiting actin depolymerization, expanding dendritic spine volume, and contributing to the induction of LTP.

This was first demonstrated in the rodent neocortex with paired recordings: near-synchronous firing produced inhibitory long-term potentiation, but if the inhibitory neuron consistently led the excitatory cell by a few milliseconds, the connection underwent lasting depression.^[7] Such timing rules for inhibitory STDP are essentially the mirror-image of classic excitatory STDP and serve to modulate circuit excitability. Strengthening an inhibitory synapse when it fires after the excitatory neuron can be interpreted as enhancing feedback inhibition, whereas weakening it when it fires before the excitatory neuron reduces feed-forward inhibition, potentially allowing the excitatory neuron to fire more easily in the future.

The physiological roles of inhibitory STDP are still being elucidated. One important function appears to be the fine-tuning of excitatory–inhibitory balance in neural networks. Timing-dependent changes at inhibitory synapses have been shown to contribute to sensory map plasticity. In the auditory cortex, for instance, plasticity of inhibitory synapses following specific spike-timing protocols has been linked to the remapping of neuronal frequency preference, implying that inhibitory STDP participates in experience-dependent reorganization of receptive fields.^[19]

Moreover, distinct molecular mechanisms underlie STDP at inhibitory synapses. Unlike excitatory LTP/LTD (often NMDA receptor-dependent), inhibitory long-term plasticity can require retrograde signaling and neuromodulators. In hippocampal CA1 pyramidal cells, an STDP protocol that induces long-term depression of GABAergic synapses was found to depend on endocannabinoid release and the activation of presynaptic M2 muscarinic acetylcholine receptors. In this case, precisely timed postsynaptic spikes trigger the synthesis of endocannabinoids which act on CB1 receptors at the interneuron terminals, reducing GABA release, a mechanism quite different from NMDA-dependent excitatory STDP.^{[20][21]} Such requirements highlight that inhibitory STDP often engages metaplastic gating factors (e.g., cholinergic tone or cannabinoid signaling) to occur.

Overall, STDP in inhibitory synapses adds an important plasticity paradigm: it enables activity-dependent adjustment of inhibition in neural circuits, complementing excitatory plasticity and ensuring that the timing of inhibition relative to excitation can be optimized through experience.



Endocannabinoids released from the postsynaptic neuron modulate synaptic strength. In homosynaptic plasticity, these effects are confined to the active synapse, whereas in heterosynaptic plasticity, the signaling can influence neighboring, inactive synapses. (See section "Interaction with Other Forms of Plasticity" for more detail)

Neuromodulation of STDP

Neuromodulators strongly influence STDP by modifying the timing rules that determine whether a synapse is strengthened or weakened. These changes depend on broader brain states, such as arousal, attention, or reward, allowing synaptic plasticity to be tuned to the behavioral context. Experimental studies have shown that the presence of neuromodulatory signals (such as dopamine, acetylcholine, noradrenaline, serotonin, etc.) can gate or alter the outcome of STDP. For example, in hippocampal and cortical slices, activation of β -adrenergic receptors can convert what would normally be an LTD-induced timing into a net LTP. One study demonstrated that applying isoproterenol (a β -adrenergic agonist) during an STDP induction protocol abolished the LTD that would occur when postsynaptic spikes followed presynaptic spikes (post-pre pairing), and instead resulted in potentiation for all spike-timing intervals tested.^[22] In essence, adrenergic neuromodulation extended the potentiation window and prevented LTD, overriding the usual temporal requirements. This finding aligns with a broader principle: neuromodulators that activate cAMP/PKA pathways (like norepinephrine via β receptors or dopamine via D1 receptors) tend to bias synaptic plasticity toward potentiation, whereas other signals can bias toward depression.^[23]

Acetylcholine and dopamine provide an example of sequential neuromodulation of STDP. Acetylcholine (acting largely through muscarinic receptors in the forebrain) can bias STDP toward depression, for instance, by lowering the threshold for LTD induction or narrowing the LTP window. This might occur during exploratory or attentive states when acetylcholine is high. If a reward or salient outcome then follows, dopamine is released and can invert the synaptic changes from depression to potentiation. In hippocampal experiments, an initial conditioning in the presence of acetylcholine resulted in synaptic weakening, but a subsequent application of dopamine converted that LTD into LTP.^[24] Moreover, dopamine can broaden the STDP time window for potentiation and can even rescue LTP if given shortly after the spike pairing, effectively imparting a reward timing signal to STDP. This modulatory sequence (ACh followed by dopamine) has been proposed as a cellular mechanism for associating sensory cues or actions with delayed reward, by first allowing synapses to depress and then reinforcing them if the outcome is positive.^[24]

In general, neuromodulators enable state-dependent plasticity: the brain can promote or prevent STDP under certain conditions. For instance, in the presence of cholinergic tone (e.g., during waking exploration), some cortical synapses may not potentiate with STDP unless dopaminergic or noradrenergic signals also indicate a behaviorally important event.^[23] Neuromodulation of STDP is not limited to the classic neurotransmitters; it potentially extends to neuropeptides, stress hormones, and other modulators that adjust neural plasticity. This dynamic regulation ensures that STDP in real neural circuits is context-sensitive. Synaptic changes occur not only based on spike timing but also in conjunction with the animal's arousal, attention, and reward environment, thereby linking synaptic plasticity to learning and motivation processes.^[23]

History

The conceptual foundation for spike-timing-dependent plasticity (STDP) is rooted in the mid-20th century work of Canadian psychologist Donald Hebb. In his 1949 book, *The Organization of Behavior*, Hebb proposed that the repeated and persistent activation of a presynaptic neuron contributes to the firing

of a postsynaptic neuron. This interaction strengthens the connection between them. This principle, now known as Hebbian theory, is often summarized as "cells that fire together, wire together".^[25] While Hebb's framework emphasized the co-occurrence of neural activity, it did not account for the precise temporal order of neuronal firing.

As early as 1973, M. M. Taylor proposed a theoretical learning rule in which synapses would be strengthened if a presynaptic spike reliably preceded a postsynaptic spike, and weakened if the timing was reversed.^[26] He suggested that such a temporally sensitive mechanism could support more efficient encoding of input patterns, although this idea did not receive widespread attention at the time.

Experimental studies in the 1980s and early 1990s began to probe the timing dependence of synaptic plasticity more directly. Notably, W. B. Levy and O. Steward (1983) demonstrated that the timing between pre- and postsynaptic activity could influence the direction of synaptic change.^[27] Subsequent work by Y. Dan and Mu-Ming Poo (1992) at neuromuscular synapses, and by D. Debanne, B. Gähwiler, and S. Thompson (1994) in the hippocampus, showed that asynchronous pairing of spikes could lead to long-term synaptic depression (LTD).^{[28][29]}

The modern concept of STDP began to take clearer shape through work in the mid-1990s. Henry Markram, working in Bert Sakmann's lab, used dual patch-clamp recordings to show that the order of spike firing between two connected neurons could bidirectionally modify synaptic strength. When the presynaptic neuron was activated approximately 10 milliseconds before the postsynaptic neuron, the connection was strengthened; reversing the order led to weakening. Although these findings were presented in abstracts and conference proceedings earlier, they were formally published in 1997.^[30]

Shortly after, in a seminal 1998 study, Guoqiang Bi and Mu-Ming Poo systematically mapped the time-dependent curve relating the order of pre- and postsynaptic spikes to the magnitude and direction of synaptic change in cultured hippocampal neurons. Their work demonstrated that small differences in spike timing—on the order of tens of milliseconds—could induce either long-term potentiation (LTP) or long-term depression (LTD).^[5] This study became a foundational reference for what is now recognized as the canonical STDP "learning window."

Since these early findings, STDP has been confirmed in a wide range of preparations, including brain slices and in vivo systems, across different species and brain regions. STDP-like mechanisms have been observed in the visual cortex, hippocampus, cerebellum, and neocortex, as well as in sensory-evoked responses in intact animals.^{[31][32]} It has also been demonstrated that the initially asymmetric timing window seen in early STDP studies can evolve over time; for example, one study showed that a more symmetric "LTP-only" profile may emerge several days after initial induction.^[33]

STDP is now widely regarded as a key biological mechanism supporting Hebbian learning, particularly during development, where it is thought to refine neural circuits based on experience.^{[34][35]}

From Hebbian rule to STDP

According to Hebb's original postulate, synaptic strength increases when the presynaptic neuron persistently contributes to the firing of the postsynaptic neuron. Conversely, synapses weaken when the activity of the presynaptic neuron is consistently unrelated to postsynaptic firing. These principles are

often simplified by the mnemonics: "cells that fire together, wire together," and "out of sync, lose their link".^[25]

However, Hebb's formulation did not explicitly address the directionality of firing. Two neurons firing at exactly the same time cannot satisfy a causal relationship. To contribute to the firing of a postsynaptic neuron, the presynaptic neuron must fire shortly before it. Experimental studies confirmed that this precise temporal relationship is critical for synaptic modification: when the presynaptic spike occurs just before the postsynaptic one, the synapse is potentiated; if it follows, the synapse is depressed.^{[30][5]}

Furthermore, STDP appears to depend not only on the timing between individual spikes, but also on the consistency with which presynaptic firing predicts postsynaptic activation. This mirrors the principle of contingency (learning theory), well known in classical conditioning: consistent predictive relationships, rather than mere co-occurrence, are necessary for robust learning.^[30]

Role in hippocampal learning

Spike-timing-dependent plasticity (STDP) operates over a narrow time window—typically on the order of tens of milliseconds—yet the hippocampus is capable of associating events that unfold over several seconds, as is required during episodic memory formation. This apparent mismatch in time scales is reconciled, in part, by the presence of theta waves (4–8 Hz), rhythmic electrical activity in the hippocampus that organizes neuronal firing during active exploration and learning.^[31]

As an animal experiences a sequence of events or traverses space, hippocampal place cells fire at progressively earlier phases of each theta cycle, a phenomenon known as phase precession. This effectively compresses temporally distributed spikes into a narrower window—aligning them within a single theta cycle (~100–150 ms)—allowing spikes from sequentially active neurons to fall within the STDP window.^[36]

In the recurrent networks of the CA3 region, this phase alignment enables the strengthening of synaptic connections between neurons representing adjacent elements of an experience. STDP can thus support the formation of ordered memory traces by selectively reinforcing connections where presynaptic activity consistently predicts postsynaptic firing.^[37] This mechanism provides a biological basis for linking discrete experiences into coherent episodes and may underlie the hippocampus's role in encoding the temporal structure of memory.^{[31][37]}

Computational models and applications

Training spiking neural networks

The principles of STDP can be utilized in the training of artificial spiking neural networks. Using this approach the weight of a connection between two neurons is increased if the time at which a presynaptic spike (t_{pre}) occurs is shortly before the time of a post synaptic spike (t_{post}), ie. $t = t_{post} - t_{pre}$ and $t > 0$. The size of the weight increase is dependent on the value of t and decreases exponentially as the value of t increases given by the equation:

$$A_+ \exp\left(\frac{-t}{\tau_+}\right)$$

where A_+ is the maximum possible change and τ_+ is the time constant.

If the opposite scenario occurs ie a post synaptic spike occurs before a presynaptic spike then the weight is instead reduced according to the equation:

$$A_- \exp\left(\frac{t}{\tau_-}\right)$$

Where A_- and τ_- serve the same function of defining the maximum possible change and time constant as before respectively.

The parameters that define the decay profile (A_+ , A_- , etc.) do not necessarily have to be fixed across the entire network and different synapses may have different shapes associated with them.

Biological evidence suggests that this pairwise STDP approach cannot give a complete description of a biological neuron and more advanced approaches which look at symmetric triplets of spikes (pre-post-pre, post-pre-post) have been developed and these are believed to be more biologically plausible.^[32]

Developmental and Critical Period Plasticity

STDP is thought to play significant roles in neural development, particularly during critical periods when activity-dependent refinement of connections is highly pronounced. In early postnatal development, synaptic plasticity mechanisms are transiently enhanced to allow experience to shape neural circuits such as sensory maps. STDP provides an elegant rule for such refinement, and evidence suggests that STDP properties change over development to support different phases of circuit maturation. In the rodent barrel cortex (whisker somatosensory map), for example, synapses initially exhibit a unimodal timing rule, essentially all correlated activity results in potentiation in the first two postnatal weeks. During this stage, any pairing of pre- and postsynaptic spikes tends to strengthen the connection, consistent with a developmental period of synapse stabilization. Then, around the onset of the critical period (approximately 15 days postnatal), classical bidirectional Hebbian STDP emerges at these synapses.^[38] The appearance of Hebbian, bidirectional STDP coincides with the maturation of local inhibitory circuits, particularly parvalbumin-positive interneurons, which sharpen the timing of postsynaptic spikes and restrict the time window for plasticity.^[39]

This developmental switch from "all-to-potentiation" to "timing-sensitive" plasticity marks a transition from a phase of general connectivity strengthening to one of competitive refinement. It is essential for the onset of the critical period in barrel cortex, as the newly emerged STDP rule allows for the fine-scale weakening of less-active inputs and strengthening of well-timed inputs, sculpting the receptive fields of layer 2/3 neurons to better discriminate whisker-specific inputs.^{[38][40]}

More broadly, STDP is implicated in critical period plasticity across multiple brain regions. During critical periods in visual cortex, for instance, the timing of pre- and postsynaptic activity (driven by visual experience) likely contributes to synaptic competition underlying ocular dominance plasticity and orientation selectivity.^{[41][42]} The flexibility of STDP appears to be greater in juvenile brains, studies have

found that younger animals can exhibit larger timing-dependent changes, whereas in mature circuits the same STDP protocols might produce smaller effects unless neuromodulatory conditions re-enable plasticity.^[41] One contributing factor is the prevalence of NR2B-containing NMDA receptors in early development, which extend the duration of synaptic currents and broaden the timing window for STDP induction.^[43] Other developmental changes, such as the strengthening of perisomatic inhibition and the formation of extracellular matrix structures (e.g., perineuronal nets), serve to tighten STDP time windows and limit plasticity, contributing to critical period closure.^{[42][44]}

Experimental manipulations confirm these roles: increasing inhibition prematurely can close a critical period and suppress STDP, while reducing inhibition or delaying circuit maturation can prolong a plastic, STDP-permissive state.^{[42][45]} STDP's contribution to developmental plasticity is also supported by theoretical models, which demonstrate that timing-based rules are sufficient to refine topographic maps and receptive fields in early sensory systems.^[46] In summary, STDP works in concert with developmental regulatory processes to ensure that early in life, synaptic connections are broadly strengthened by correlated activity, and later, as inhibition matures, STDP becomes more selective, enabling critical period experiences to fine-tune synapses and solidify mature circuit properties.

See also

- [Synaptic plasticity](#)
- [Didactic organisation](#)

References

1. Bi, Guo-qiang; Poo, Mu-ming (1998). "Synaptic Modifications in Cultured Hippocampal Neurons: Dependence on Spike Timing, Synaptic Strength, and Postsynaptic Cell Type" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6793365>). *The Journal of Neuroscience*. **18** (24): 10464–10472. doi:10.1523/JNEUROSCI.18-24-10464.1998 (<https://doi.org/10.1523%2FJNEUROSCI.18-24-10464.1998>). ISSN 0270-6474 (<https://search.worldcat.org/issn/0270-6474>). PMC 6793365 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6793365>). PMID 9852584 (<https://pubmed.ncbi.nlm.nih.gov/9852584>).
2. Caporale, Natalia; Dan, Yang (2008). "Spike Timing–Dependent Plasticity: A Hebbian Learning Rule" (<https://www.annualreviews.org/content/journals/10.1146/annurev.neuro.31.060407.125639>). *Annual Review of Neuroscience*. **31** (1): 25–46. doi:10.1146/annurev.neuro.31.060407.125639 (<https://doi.org/10.1146%2Fannurev.neuro.31.060407.125639>). ISSN 0147-006X (<https://search.worldcat.org/issn/0147-006X>). PMID 18275283 (<https://pubmed.ncbi.nlm.nih.gov/18275283>).
3. Dan, Yang; Poo, Mu-Ming (2006). "Spike Timing-Dependent Plasticity: From Synapse to Perception" (<https://www.physiology.org/doi/10.1152/physrev.00030.2005>). *Physiological Reviews*. **86** (3): 1033–1048. doi:10.1152/physrev.00030.2005 (<https://doi.org/10.1152%2Fphysrev.00030.2005>). ISSN 0031-9333 (<https://search.worldcat.org/issn/0031-9333>). PMID 16816145 (<https://pubmed.ncbi.nlm.nih.gov/16816145>).
4. Feldman, Daniel E. (2012). "The Spike-Timing Dependence of Plasticity" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3431193>). *Neuron*. **75** (4): 556–571. doi:10.1016/j.neuron.2012.08.001 (<https://doi.org/10.1016%2Fj.neuron.2012.08.001>). PMC 3431193 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3431193>). PMID 22920249 (<https://pubmed.ncbi.nlm.nih.gov/22920249>).

5. Bi, Guo-qiang; Poo, Mu-ming (1998). "Synaptic Modifications in Cultured Hippocampal Neurons: Dependence on Spike Timing, Synaptic Strength, and Postsynaptic Cell Type" (<https://www.jneurosci.org/content/18/24/10464>). *Journal of Neuroscience*. **18** (24): 10464–10472. doi:10.1523/JNEUROSCI.18-24-10464.1998 (<https://doi.org/10.1523%2FJNEUROSCI.18-24-10464.1998>). ISSN 0270-6474 (<https://search.worldcat.org/issn/0270-6474>). PMC 6793365 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6793365>). PMID 9852584 (<https://pubmed.ncbi.nlm.nih.gov/9852584>).
6. Brzosko, Zuzanna; Mierau, Susanna B.; Paulsen, Ole (2019-08-21). "Neuromodulation of Spike-Timing-Dependent Plasticity: Past, Present, and Future". *Neuron*. **103** (4): 563–581. doi:10.1016/j.neuron.2019.05.041 (<https://doi.org/10.1016%2Fj.neuron.2019.05.041>). ISSN 1097-4199 (<https://search.worldcat.org/issn/1097-4199>). PMID 31437453 (<https://pubmed.ncbi.nlm.nih.gov/31437453>).
7. Capogna, Marco; Castillo, Pablo E.; Maffei, Arianna (2021). "The ins and outs of inhibitory synaptic plasticity: Neuron types, molecular mechanisms and functional roles" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9150441>). *European Journal of Neuroscience*. **54** (8): 6882–6901. doi:10.1111/ejn.14907 (<https://doi.org/10.1111%2Fejn.14907>). ISSN 0953-816X (<https://search.worldcat.org/issn/0953-816X>). PMC 9150441 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9150441>). PMID 32663353 (<https://pubmed.ncbi.nlm.nih.gov/32663353>).
8. Sjöström, Per Jesper; Turrigiano, Gina G.; Nelson, Sacha B. (2001-12-20). "Rate, Timing, and Cooperativity Jointly Determine Cortical Synaptic Plasticity" (<https://doi.org/10.1016%2FS0896-6273%2801%2900542-6>). *Neuron*. **32** (6): 1149–1164. doi:10.1016/S0896-6273(01)00542-6 (<https://doi.org/10.1016%2FS0896-6273%2801%2900542-6>). ISSN 0896-6273 (<https://search.worldcat.org/issn/0896-6273>). PMID 11754844 (<https://pubmed.ncbi.nlm.nih.gov/11754844>).
9. Turrigiano, G. (2011-11-15). "Homeostatic Synaptic Plasticity: Local and Global Mechanisms for Stabilizing Neuronal Function" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3249629>). *Cold Spring Harbor Perspectives in Biology*. **4** (1) a005736. doi:10.1101/cshperspect.a005736 (<https://doi.org/10.1101%2Fcshperspect.a005736>). ISSN 1943-0264 (<https://search.worldcat.org/issn/1943-0264>). PMC 3249629 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3249629>). PMID 22086977 (<https://pubmed.ncbi.nlm.nih.gov/22086977>).
10. Chistiakova, Marina; Bannion, Nicholas M.; Chen, Jen-Yung; Bazhenov, Maxim; Volgushev, Maxim (2015-07-13). "Homeostatic role of heterosynaptic plasticity: models and experiments" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4500102>). *Frontiers in Computational Neuroscience*. **9**: 89. doi:10.3389/fncom.2015.00089 (<https://doi.org/10.3389%2Ffncom.2015.00089>). ISSN 1662-5188 (<https://search.worldcat.org/issn/1662-5188>). PMC 4500102 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4500102>). PMID 26217218 (<https://pubmed.ncbi.nlm.nih.gov/26217218>).
11. Garad, Machhindra; Edelmann, Elke; Leßmann, Volkmar (2021-01-30). "Impairment of Spike-Timing-Dependent Plasticity at Schaffer Collateral-CA1 Synapses in Adult APP/PS1 Mice Depends on Proximity of A β Plaques" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7866519>). *International Journal of Molecular Sciences*. **22** (3): 1378. doi:10.3390/ijms22031378 (<https://doi.org/10.3390%2Fijms22031378>). ISSN 1422-0067 (<https://search.worldcat.org/issn/1422-0067>). PMC 7866519 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7866519>). PMID 33573114 (<https://pubmed.ncbi.nlm.nih.gov/33573114>).
12. Meredith, Rhiannon M.; Mansvelder, Huibert D. (2010-06-10). "STDP and mental retardation: dysregulation of dendritic excitability in Fragile X syndrome" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3059693>). *Frontiers in Synaptic Neuroscience*. **2**: 10. doi:10.3389/fnsyn.2010.00010 (<https://doi.org/10.3389%2Ffnsyn.2010.00010>). ISSN 1663-3563 (<https://search.worldcat.org/issn/1663-3563>). PMC 3059693 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3059693>). PMID 21423496 (<https://pubmed.ncbi.nlm.nih.gov/21423496>).

13. Calabresi, Paolo; Picconi, Barbara; Tozzi, Alessandro; Ghiglieri, Veronica; Di Filippo, Massimiliano (2014). "Direct and indirect pathways of basal ganglia: a critical reappraisal" (<https://www.nature.com/articles/nn.3743>). *Nature Neuroscience*. **17** (8): 1022–1030. doi:10.1038/nn.3743 (<https://doi.org/10.1038%2Fnn.3743>). ISSN 1546-1726 (<https://search.worldcat.org/issn/1546-1726>). PMID 25065439 (<https://pubmed.ncbi.nlm.nih.gov/25065439>).
14. Bernard, C.; Cossart, R.; Hirsch, J. C.; Esclapez, M.; Ben-Ari, Y. (2000). "What is GABAergic inhibition? How is it modified in epilepsy?". *Epilepsia*. **41** Suppl 6: S90–95. doi:10.1111/j.1528-1157.2000.tb01564.x (<https://doi.org/10.1111%2Fj.1528-1157.2000.tb01564.x>). ISSN 0013-9580 (<https://search.worldcat.org/issn/0013-9580>). PMID 10999527 (<https://pubmed.ncbi.nlm.nih.gov/10999527>).
15. Magee, Jeffrey C.; Grienberger, Christine (2020). "Synaptic Plasticity Forms and Functions" (<https://www.annualreviews.org/content/journals/10.1146/annurev-neuro-090919-022842>). *Annual Review of Neuroscience*. **43**: 95–117. doi:10.1146/annurev-neuro-090919-022842 (<https://doi.org/10.1146%2Fannurev-neuro-090919-022842>). ISSN 0147-006X (<https://search.worldcat.org/issn/0147-006X>). PMID 32075520 (<https://pubmed.ncbi.nlm.nih.gov/32075520>).
16. Lisman, John; Cooper, Katherine; Sehgal, Megha; Silva, Alcino J. (2018). "Memory formation depends on both synapse-specific modifications of synaptic strength and cell-specific increases in excitability" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5915620>). *Nature Neuroscience*. **21** (3): 309–314. doi:10.1038/s41593-018-0076-6 (<https://doi.org/10.1038%2Fs41593-018-0076-6>). ISSN 1546-1726 (<https://search.worldcat.org/issn/1546-1726>). PMC 5915620 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5915620>). PMID 29434376 (<https://pubmed.ncbi.nlm.nih.gov/29434376>).
17. Morrison, Abigail; Diesmann, Markus; Gerstner, Wulfram (2008). "Phenomenological models of synaptic plasticity based on spike timing" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2799003>). *Biological Cybernetics*. **98** (6): 459–478. doi:10.1007/s00422-008-0233-1 (<https://doi.org/10.1007%2Fs00422-008-0233-1>). ISSN 1432-0770 (<https://search.worldcat.org/issn/1432-0770>). PMC 2799003 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2799003>). PMID 18491160 (<https://pubmed.ncbi.nlm.nih.gov/18491160>).
18. Vickers, Evan D.; Clark, Christopher; Osypenko, Denys; Fratzl, Alex; Kochubey, Olexiy; Bettler, Bernhard; Schneggenburger, Ralf (2018). "Parvalbumin-Interneuron Output Synapses Show Spike-Timing-Dependent Plasticity that Contributes to Auditory Map Remodeling" (<https://www.sciencedirect.com/science/article/pii/S0896627318305920>). *Neuron*. **99** (4): 720–735.e6. doi:10.1016/j.neuron.2018.07.018 (<https://doi.org/10.1016%2Fj.neuron.2018.07.018>). ISSN 0896-6273 (<https://search.worldcat.org/issn/0896-6273>). PMID 30078579 (<https://pubmed.ncbi.nlm.nih.gov/30078579>).
19. Tremblay, Robin; Lee, Soohyun; Rudy, Bernardo (2016). "GABAergic Interneurons in the Neocortex: From Cellular Properties to Circuits" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4980915>). *Neuron*. **91** (2): 260–292. doi:10.1016/j.neuron.2016.06.033 (<https://doi.org/10.1016%2Fj.neuron.2016.06.033>). ISSN 0896-6273 (<https://search.worldcat.org/issn/0896-6273>). PMC 4980915 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4980915>). PMID 27477017 (<https://pubmed.ncbi.nlm.nih.gov/27477017>).
20. Kim, Jimok; Isokawa, Masako; Ledent, Catherine; Alger, Bradley E. (2002). "Activation of Muscarinic Acetylcholine Receptors Enhances the Release of Endogenous Cannabinoids in the Hippocampus" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6758770>). *The Journal of Neuroscience*. **22** (23): 10182–10191. doi:10.1523/JNEUROSCI.22-23-10182.2002 (<https://doi.org/10.1523%2FJNEUROSCI.22-23-10182.2002>). ISSN 0270-6474 (<https://search.worldcat.org/issn/0270-6474>). PMC 6758770 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6758770>). PMID 12451119 (<https://pubmed.ncbi.nlm.nih.gov/12451119>).

21. Pitler, T. A.; Alger, B. E. (1992). "Postsynaptic spike firing reduces synaptic GABA responses in hippocampal pyramidal cells" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6575956>). *The Journal of Neuroscience*. **12** (10): 4122–4132. doi:10.1523/JNEUROSCI.12-10-04122.1992 (<https://doi.org/10.1523%2FJNEUROSCI.12-10-04122.1992>). ISSN 0270-6474 (<https://search.worldcat.org/issn/0270-6474>). PMC 6575956 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6575956>). PMID 1403103 (<https://pubmed.ncbi.nlm.nih.gov/1403103>).
22. Seol, Geun Hee; Ziburkus, Jokubas; Huang, ShiYong; Song, Lihua; Kim, In Tae; Takamiya, Kogo; Huganir, Richard L.; Lee, Hey-Kyoung; Kirkwood, Alfredo (2007). "Neuromodulators Control the Polarity of Spike-Timing-Dependent Synaptic Plasticity" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2756178>). *Neuron*. **55** (6): 919–929. doi:10.1016/j.neuron.2007.08.013 (<https://doi.org/10.1016%2Fj.neuron.2007.08.013>). PMC 2756178 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2756178>). PMID 17880895 (<https://pubmed.ncbi.nlm.nih.gov/17880895>).
23. Frémaux, Nicolas; Gerstner, Wulfram (2016). "Neuromodulated Spike-Timing-Dependent Plasticity, and Theory of Three-Factor Learning Rules" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4717313>). *Frontiers in Neural Circuits*. **9**: 85. doi:10.3389/fncir.2015.00085 (<https://doi.org/10.3389%2Ffncir.2015.00085>). ISSN 1662-5110 (<https://search.worldcat.org/issn/1662-5110>). PMC 4717313 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4717313>). PMID 26834568 (<https://pubmed.ncbi.nlm.nih.gov/26834568>).
24. Brzosko, Zuzanna; Zannone, Sara; Schultz, Wolfram; Clopath, Claudia; Paulsen, Ole (2017). Bartos, Marlene (ed.). "Sequential neuromodulation of Hebbian plasticity offers mechanism for effective reward-based navigation" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5546805>). *eLife*. **6** e27756. doi:10.7554/eLife.27756 (<https://doi.org/10.7554%2FeLife.27756>). ISSN 2050-084X (<https://search.worldcat.org/issn/2050-084X>). PMC 5546805 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5546805>). PMID 28691903 (<https://pubmed.ncbi.nlm.nih.gov/28691903>).
25. Morris, R. G. M (1999). "D.O. Hebb: The Organization of Behavior, Wiley: New York; 1949" (<https://linkinghub.elsevier.com/retrieve/pii/S0361923099001823>). *Brain Research Bulletin*. **50** (5): 437. doi:10.1016/S0361-9230(99)00182-3 (<https://doi.org/10.1016%2FS0361-9230%2899%2900182-3>). ISSN 0361-9230 (<https://search.worldcat.org/issn/0361-9230>). PMID 10643472 (<https://pubmed.ncbi.nlm.nih.gov/10643472>).
26. Taylor MM (1973). "The Problem of Stimulus Structure in the Behavioural Theory of Perception" (http://www.mmtaylor.net/Academic/SouthAfricanJ_OCR_edit.doc). *South African Journal of Psychology*. **3**: 23–45.
27. Levy, W. B.; Steward, O. (1983). "Temporal contiguity requirements for long-term associative potentiation/depression in the hippocampus" (<https://linkinghub.elsevier.com/retrieve/pii/0306452283900106>). *Neuroscience*. **8** (4): 791–797. doi:10.1016/0306-4522(83)90010-6 (<https://doi.org/10.1016%2F0306-4522%2883%2990010-6>). ISSN 0306-4522 (<https://search.worldcat.org/issn/0306-4522>). PMID 6306504 (<https://pubmed.ncbi.nlm.nih.gov/6306504>).
28. Dan, Yang; Poo, Mu-ming (1992). "Hebbian Depression of Isolated Neuromuscular Synapses in Vitro" (<https://www.science.org/doi/10.1126/science.1317971>). *Science*. **256** (5063): 1570–1573. Bibcode:1992Sci...256.1570D (<https://ui.adsabs.harvard.edu/abs/1992Sci...256.1570D>). doi:10.1126/science.1317971 (<https://doi.org/10.1126%2Fscience.1317971>). ISSN 0036-8075 (<https://search.worldcat.org/issn/0036-8075>). PMID 1317971 (<https://pubmed.ncbi.nlm.nih.gov/1317971>).
29. Debanne, D; Gähwiler, B H; Thompson, S M (1994). "Asynchronous pre- and postsynaptic activity induces associative long-term depression in area CA1 of the rat hippocampus in vitro" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC521471>). *Proceedings of the National Academy of Sciences*. **91** (3): 1148–1152. Bibcode:1994PNAS...91.1148D (<https://ui.adsabs.harvard.edu/abs/1994PNAS...91.1148D>). doi:10.1073/pnas.91.3.1148 (<https://doi.org/10.1073%2Fpnas.91.3.1148>). PMC 521471 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC521471>). PMID 7905631 (<https://pubmed.ncbi.nlm.nih.gov/7905631>).

30. Markram, Henry; Lübke, Joachim; Frotscher, Michael; Sakmann, Bert (1997). "Regulation of Synaptic Efficacy by Coincidence of Postsynaptic APs and EPSPs" (<https://www.science.org/doi/10.1126/science.275.5297.213>). *Science*. **275** (5297): 213–215. doi:10.1126/science.275.5297.213 (<https://doi.org/10.1126%2Fscience.275.5297.213>). PMID 8985014 (<https://pubmed.ncbi.nlm.nih.gov/8985014>).
31. Kovács KA (2020). "Episodic Memories: How do the Hippocampus and the Entorhinal Ring Attractors Cooperate to Create Them?" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7511719>). *Frontiers in Systems Neuroscience*. **14** 559168: 68. doi:10.3389/fnsys.2020.559186 (<https://doi.org/10.3389%2Ffnsys.2020.559186>). PMC 7511719 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7511719>). PMID 33013334 (<https://pubmed.ncbi.nlm.nih.gov/33013334>).
32. Taherkhani, Aboozar; Belatreche, Ammar; Li, Yuhua; Cosma, Georgina; Maguire, Liam P.; McGinnity, T. M. (2020). "A review of learning in biologically plausible spiking neural networks" (<https://www.sciencedirect.com/science/article/pii/S0893608019303181>). *Neural Networks*. **122**: 253–272. doi:10.1016/j.neunet.2019.09.036 (<https://doi.org/10.1016%2Fj.neunet.2019.09.036>). ISSN 0893-6080 (<https://search.worldcat.org/issn/0893-6080>). PMID 31726331 (<https://pubmed.ncbi.nlm.nih.gov/31726331>).
33. Wittenberg, G. M.; Wang, S. S.- H. (2006). "Malleability of Spike-Timing-Dependent Plasticity at the CA3-CA1 Synapse" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6674029>). *Journal of Neuroscience*. **26** (24): 6610–6617. doi:10.1523/JNEUROSCI.5388-05.2006 (<https://doi.org/10.1523%2FJNEUROSCI.5388-05.2006>). ISSN 0270-6474 (<https://search.worldcat.org/issn/0270-6474>). PMC 6674029 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6674029>). PMID 16775149 (<https://pubmed.ncbi.nlm.nih.gov/16775149>).
34. Kirkwood, Alfredo; Rioult, Marika G.; Bear, Mark F. (1996). "Experience-dependent modification of synaptic plasticity in visual cortex" (<https://www.nature.com/articles/381526a0>). *Nature*. **381** (6582): 526–528. Bibcode:1996Natur.381..526K (<https://ui.adsabs.harvard.edu/abs/1996Natur.381..526K>). doi:10.1038/381526a0 (<https://doi.org/10.1038%2F381526a0>). ISSN 1476-4687 (<https://search.worldcat.org/issn/1476-4687>). PMID 8632826 (<https://pubmed.ncbi.nlm.nih.gov/8632826>).
35. Song, Sen; Miller, Kenneth D.; Abbott, L. F. (2000). "Competitive Hebbian learning through spike-timing-dependent synaptic plasticity" (https://www.nature.com/articles/nn0900_919). *Nature Neuroscience*. **3** (9): 919–926. doi:10.1038/78829 (<https://doi.org/10.1038%2F78829>). ISSN 1546-1726 (<https://search.worldcat.org/issn/1546-1726>). PMID 10966623 (<https://pubmed.ncbi.nlm.nih.gov/10966623>).
36. Skaggs, William E.; McNaughton, Bruce L.; Wilson, Matthew A.; Barnes, Carol A. (1996). "Theta phase precession in hippocampal neuronal populations and the compression of temporal sequences" ([https://onlinelibrary.wiley.com/doi/10.1002/\(SICI\)1098-1063\(1996\)6:2<149::AID-HIPO6>3.0.CO;2-K](https://onlinelibrary.wiley.com/doi/10.1002/(SICI)1098-1063(1996)6:2<149::AID-HIPO6>3.0.CO;2-K)). *Hippocampus*. **6** (2): 149–172. doi:10.1002/(SICI)1098-1063(1996)6:2<149::AID-HIPO6>3.0.CO;2-K ([https://doi.org/10.1002%2F%28SICI%291098-1063\(1996\)6:2%2C149::AID-HIPO6%3A2%3C149%3A%3AAID-HIPO6%3E3.0.CO%3B2-K](https://doi.org/10.1002%2F%28SICI%291098-1063(1996)6:2%2C149::AID-HIPO6%3A2%3C149%3A%3AAID-HIPO6%3E3.0.CO%3B2-K)). ISSN 1050-9631 (<https://search.worldcat.org/issn/1050-9631>). PMID 8797016 (<https://pubmed.ncbi.nlm.nih.gov/8797016>).
37. Mehta, Mayank R.; Quirk, Michael C.; Wilson, Matthew A. (2000). "Experience-Dependent Asymmetric Shape of Hippocampal Receptive Fields" (<https://linkinghub.elsevier.com/retrieve/pii/S0896627300810727>). *Neuron*. **25** (3): 707–715. doi:10.1016/S0896-6273(00)81072-7 (<https://doi.org/10.1016%2FS0896-6273%2800%2981072-7>). ISSN 0896-6273 (<https://search.worldcat.org/issn/0896-6273>). PMID 10774737 (<https://pubmed.ncbi.nlm.nih.gov/10774737>).

38. Itami, Chiaki; Kimura, Fumitaka (2012). "Developmental Switch in Spike Timing-Dependent Plasticity at Layers 4–2/3 in the Rodent Barrel Cortex" (<https://www.jneurosci.org/content/32/43/15000>). *Journal of Neuroscience*. **32** (43): 15000–15011. doi:10.1523/JNEUROSCI.2506-12.2012 (<https://doi.org/10.1523%2FJNEUROSCI.2506-12.2012>). ISSN 0270-6474 (<https://search.worldcat.org/issn/0270-6474>). PMC 6704840 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6704840>). PMID 23100422 (<https://pubmed.ncbi.nlm.nih.gov/23100422>).
39. Hensch, Takao K. (2005). "Critical period plasticity in local cortical circuits" (<https://www.nature.com/articles/nrn1787>). *Nature Reviews Neuroscience*. **6** (11): 877–888. doi:10.1038/nrn1787 (<https://doi.org/10.1038%2Fnrn1787>). hdl:10533/174307 (<https://hdl.handle.net/10533%2F174307>). ISSN 1471-0048 (<https://search.worldcat.org/issn/1471-0048>). PMID 16261181 (<https://pubmed.ncbi.nlm.nih.gov/16261181>).
40. Smith, Gordon B.; Sederberg, Audrey; Elyada, Yishai M.; Van Hooser, Stephen D.; Kaschube, Matthias; Fitzpatrick, David (2015). "The development of cortical circuits for motion discrimination" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4334116>). *Nature Neuroscience*. **18** (2): 252–261. doi:10.1038/nn.3921 (<https://doi.org/10.1038%2Fnn.3921>). ISSN 1546-1726 (<https://search.worldcat.org/issn/1546-1726>). PMC 4334116 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4334116>). PMID 25599224 (<https://pubmed.ncbi.nlm.nih.gov/25599224>).
41. Feldman, Daniel E. (2009). "Synaptic mechanisms for plasticity in neocortex" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3071739>). *Annual Review of Neuroscience*. **32**: 33–55. doi:10.1146/annurev.neuro.051508.135516 (<https://doi.org/10.1146%2Fannurev.neuro.051508.135516>). ISSN 1545-4126 (<https://search.worldcat.org/issn/1545-4126>). PMC 3071739 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3071739>). PMID 19400721 (<https://pubmed.ncbi.nlm.nih.gov/19400721>).
42. Takesian, Anne E.; Hensch, Takao K. (2013). "Balancing plasticity/stability across brain development" (<https://pubmed.ncbi.nlm.nih.gov/24309249>). *Changing Brains - Applying Brain Plasticity to Advance and Recover Human Ability*. Progress in Brain Research. Vol. 207. pp. 3–34. doi:10.1016/B978-0-444-63327-9.00001-1 (<https://doi.org/10.1016%2FB978-0-444-63327-9.00001-1>). ISBN 978-0-444-63327-9. ISSN 1875-7855 (<https://search.worldcat.org/issn/1875-7855>). PMID 24309249 (<https://pubmed.ncbi.nlm.nih.gov/24309249>).
43. Philpot, B. D.; Sekhar, A. K.; Shouval, H. Z.; Bear, M. F. (2001). "Visual experience and deprivation bidirectionally modify the composition and function of NMDA receptors in visual cortex" (<https://doi.org/10.1016%2Fs0896-6273%2801%2900187-8>). *Neuron*. **29** (1): 157–169. doi:10.1016/s0896-6273(01)00187-8 (<https://doi.org/10.1016%2Fs0896-6273%2801%2900187-8>). ISSN 0896-6273 (<https://search.worldcat.org/issn/0896-6273>). PMID 11182088 (<https://pubmed.ncbi.nlm.nih.gov/11182088>).
44. Pizzorusso, Tommaso; Medini, Paolo; Berardi, Nicoletta; Chierzi, Sabrina; Fawcett, James W.; Maffei, Lamberto (2002). "Reactivation of ocular dominance plasticity in the adult visual cortex". *Science*. **298** (5596): 1248–1251. Bibcode:2002Sci...298.1248P (<https://ui.adsabs.harvard.edu/abs/2002Sci...298.1248P>). doi:10.1126/science.1072699 (<https://doi.org/10.1126%2Fscience.1072699>). ISSN 1095-9203 (<https://search.worldcat.org/issn/1095-9203>). PMID 12424383 (<https://pubmed.ncbi.nlm.nih.gov/12424383>).
45. Hensch, T. K.; Fagiolini, M.; Mataga, N.; Stryker, M. P.; Baekkeskov, S.; Kash, S. F. (1998-11-20). "Local GABA circuit control of experience-dependent plasticity in developing visual cortex" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2851625>). *Science*. **282** (5393): 1504–1508. doi:10.1126/science.282.5393.1504 (<https://doi.org/10.1126%2Fscience.282.5393.1504>). ISSN 0036-8075 (<https://search.worldcat.org/issn/0036-8075>). PMC 2851625 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2851625>). PMID 9822384 (<https://pubmed.ncbi.nlm.nih.gov/9822384>).

46. Clopath, Claudia; Büsing, Lars; Vasilaki, Eleni; Gerstner, Wulfram (2010). "Connectivity reflects coding: a model of voltage-based STDP with homeostasis" (<https://www.nature.com/articles/nn.2479>). *Nature Neuroscience*. **13** (3): 344–352. doi:10.1038/nn.2479 (<https://doi.org/10.1038%2Fnn.2479>). hdl:10044/1/21440 (<https://hdl.handle.net/10044%2F1%2F21440>). ISSN 1546-1726 (<https://search.worldcat.org/issn/1546-1726>). PMID 20098420 (<https://pubmed.ncbi.nlm.nih.gov/20098420>).

Further reading

- Rumsey CC, Abbott LF (May 2004). "Equalization of synaptic efficacy by activity- and timing-dependent synaptic plasticity". *Journal of Neurophysiology*. **91** (5): 2273–80. CiteSeerX 10.1.1.323.8776 (<https://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.323.8776>). doi:10.1152/jn.00900.2003 (<https://doi.org/10.1152%2Fjn.00900.2003>). PMID 14681332 (<https://pubmed.ncbi.nlm.nih.gov/14681332>).
- Debanne D, Gähwiler BH, Thompson SM (February 1994). "Asynchronous pre- and postsynaptic activity induces associative long-term depression in area CA1 of the rat hippocampus *in vitro*" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC521471>). *Proceedings of the National Academy of Sciences of the United States of America*. **91** (3): 1148–52. Bibcode:1994PNAS...91.1148D (<https://ui.adsabs.harvard.edu/abs/1994PNAS...91.1148D>). doi:10.1073/pnas.91.3.1148 (<https://doi.org/10.1073%2Fpnas.91.3.1148>). PMC 521471 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC521471>). PMID 7905631 (<https://pubmed.ncbi.nlm.nih.gov/7905631>).
- Sjöström PJ, Turrigiano GG, Nelson SB (December 2001). "Rate, timing, and cooperativity jointly determine cortical synaptic plasticity" (<https://doi.org/10.1016%2FS0896-6273%2801%2900542-6>). *Neuron*. **32** (6): 1149–64. doi:10.1016/S0896-6273(01)00542-6 (<https://doi.org/10.1016%2FS0896-6273%2801%2900542-6>). PMID 11754844 (<https://pubmed.ncbi.nlm.nih.gov/11754844>). S2CID 14092591 (<https://api.semanticscholar.org/CorpusID:14092591>).
- Senn W, Markram H, Tsodyks M (January 2001). "An algorithm for modifying neurotransmitter release probability based on pre- and postsynaptic spike timing". *Neural Computation*. **13** (1): 35–67. doi:10.1162/089976601300014628 (<https://doi.org/10.1162%2F089976601300014628>). PMID 11177427 (<https://pubmed.ncbi.nlm.nih.gov/11177427>). S2CID 9091640 (<https://api.semanticscholar.org/CorpusID:9091640>).
- Roberts PD, Bell CC (December 2002). "Spike timing dependent synaptic plasticity in biological systems". *Biological Cybernetics*. **87** (5–6): 392–403. doi:10.1007/s00422-002-0361-y (<https://doi.org/10.1007%2Fs00422-002-0361-y>). PMID 12461629 (<https://pubmed.ncbi.nlm.nih.gov/12461629>). S2CID 8202648 (<https://api.semanticscholar.org/CorpusID:8202648>).
- Chechik G (July 2003). "Spike-timing-dependent plasticity and relevant mutual information maximization". *Neural Computation*. **15** (7): 1481–510. CiteSeerX 10.1.1.118.4572 (<https://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.118.4572>). doi:10.1162/089976603321891774 (<https://doi.org/10.1162%2F089976603321891774>). PMID 12816563 (<https://pubmed.ncbi.nlm.nih.gov/12816563>). S2CID 10483746 (<https://api.semanticscholar.org/CorpusID:10483746>).
- Lisman J, Spruston N (July 2005). "Postsynaptic depolarization requirements for LTP and LTD: a critique of spike timing-dependent plasticity". *Nature Neuroscience*. **8** (7): 839–41. doi:10.1038/nn0705-839 (<https://doi.org/10.1038%2Fnn0705-839>). PMID 16136666 (<https://pubmed.ncbi.nlm.nih.gov/16136666>). S2CID 35555400 (<https://api.semanticscholar.org/CorpusID:35555400>).
- Sjöström, Jesper; Wulfram Gerstner (2010). "Spike-timing dependent plasticity" (<https://doi.org/10.4249%2Fscholarpedia.1362>). *Scholarpedia*. **5** (2): 1362. Bibcode:2010SchpJ...5.1362S (<https://ui.adsabs.harvard.edu/abs/2010SchpJ...5.1362S>).

doi:10.4249/scholarpedia.1362 (<https://doi.org/10.4249%2Fscholarpedia.1362>). ISSN 1941-6016 (<https://search.worldcat.org/issn/1941-6016>).

- Caporale, Natalia; Yang Dan (2008). "Spike Timing–Dependent Plasticity: A Hebbian Learning Rule". *Annual Review of Neuroscience*. **31** (1): 25–46. doi:10.1146/annurev.neuro.31.060407.125639 (<https://doi.org/10.1146%2Fannurev.neuro.31.060407.125639>). ISSN 0147-006X (<https://search.worldcat.org/issn/0147-006X>). PMID 18275283 (<https://pubmed.ncbi.nlm.nih.gov/18275283>).
- Oertner, Thomas G. (2009). "How do synapses measure milliseconds?" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2692454>). *Frontiers in Computational Neuroscience*. **3**: 7. doi:10.3389/neuro.10.007.2009 (<https://doi.org/10.3389%2Fneuro.10.007.2009>). PMC 2692454 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2692454>). PMID 19564894 (<https://pubmed.ncbi.nlm.nih.gov/19564894>).

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